Coincent Internship

(Data Science Domain)

(Feb-march 2024)

Project Report

On

“Hate speech Detection”

Submitted by : Komal Agrawal

❖ ***Abstract***

Automated hate speech detection is an important tool in combating the spread of hate speech, particularly in social media. Numerous methods have been developed for the task, including a recent proliferation of deep-learning based approaches.

A lot of methods have already been created for the automation of hate speech detection online. There are two elements to this process: identifying the qualities that these terms utilize to target a certain group and classifying textual material as hate or non-hate speech. Due to time restraints, research efforts are initiated on the latter issue in this project. For this reason, detecting hate speech is a more challenging endeavor, as our research of the language used in typical datasets reveals that hate speech lacks distinctive, discriminatory characteristics. Deep neural network topologies are very useful for capturing the meaning of hate speech and are thus proposed as feature extractors. Data from social media sites such as Twitter are used to test the effectiveness of these procedures, and they reveal a 6 percentage point improvement in macro-average F1 or a 9 percent improvement for content that has been labeled as hateful, respectively.

❖ ***Objective***

This aims to classify textual content into non-hate or hate speech, in which case the method may also identify the targeting characteristics (i.e., types of hate, such as race, and religion) in the hate speech.

In this type of project we actually learn about the hate speech.In which we detect hate speech,offensive language , normal.

❖ ***INDEX***

➢Introduction

Hate speech is one of the serious issues we see on social media platforms like Twitter and Facebook daily. There is no legal definition of hate speech because people's opinions cannot easily be classified as hateful or offensive. Nevertheless, the United Nations defines hate speech as any type of verbal, written or behavioural communication that can attack or use discriminatory language regarding a person or a group of people based on their identity based on religion, ethnicity, nationality, race, color, ancestry, gender or any other identity factor.

In Social media platforms, there are uncontrollable number of comments and posts issued every second which make it impossible to trace or control the content of such platform. Therefore, social platforms are facing a problem in limiting these posts while balancing the freedom of speech.

The problem of hate speech in social networks is technically considered as unstructured text Problem. Therefore, extracting insights and pattern from such text can be a bit challenging, owing to the context-development interpretation of natural language. Text mining technologies have the capabilities to handle the ambiguity and variability of unstructured data. Social media platforms need to detect hate speech and prevent it from going viral or ban it at the right time. So in this project, I will walk you through the task of hate speech detection with machine learning using the Python programming language.

➢Software: -

i. Anaconda navigator
ii. Jupyter or Goggle collab
iii. Python 3.0 or higher

➢Aims & Objective

This aims to classify textual content into non-offensive, less-offensive, and more-offensive.

The proposed solutions employed the different feature engineering techniques and ML algorithms to classify content as hate speech.

- ❖ The main objective of this work is to develop an automated machine learning based approach for detecting hate speech and offensive language.
- ❖ Automated detection corresponds to automated learning such as machine learning: supervised and unsupervised learning. We use a supervised learning method to detect hate and offensive language.

Classify texts into three categories based on text sentiment and other features that a text demonstrate

➢Data Collection

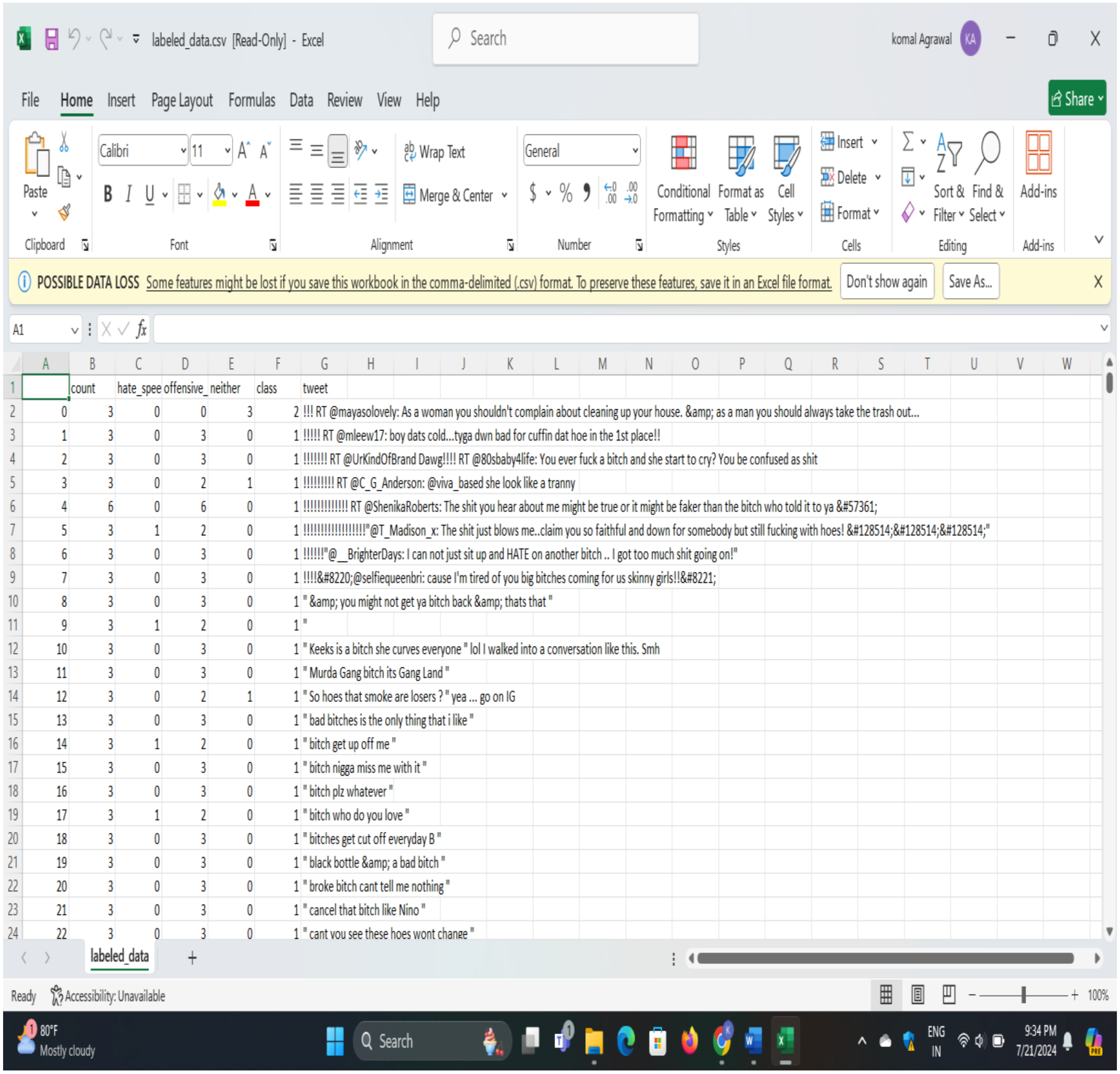
In order to build intelligent applications capable of understanding, machine learning models need to digest large amounts of structured training data. Gathering sufficient training data is the first step in solving any AI-based machine learning problem.

Data collection means pooling data by scraping, capturing, and loading it from multiple sources, including offline and online sources. High volumes of data collection or data creation can be the hardest part of a machine learning project, especially at scale. Furthermore, all datasets have flaws. This is why data preparation is so crucial in the machine learning process.

In a word, data preparation is a series of processes for making your dataset more machine learning-friendly. In a broader sense, data preparation also entails determining the best data collection mechanism. And these techniques take up the majority of machine learning time. It can take months for the first algorithm to be constructed.

In this project dataset is collected in two different ways:-

- Twitter (labeled_data_csv) Dataset is downloaded from Kaggle.
- Dataset is prepared by videos annotation



	count	hate_spee	offensive_	neither	class	tweet
0	3	0	0	3	2	!!! RT @mayasolovely: As a woman you shouldn't complain about cleaning up your house. & as a man you should always take the trash out...
1	3	0	3	0	1	!!!!! RT @mleew17: boy dats cold...tyga dwn bad for cuffin dat hoe in the 1st place!!
2	3	0	3	0	1	!!!!!!! RT @UrKindOfBrand Dawg!!!! RT @80sbaby4life: You ever fuck a bitch and she start to cry? You be confused as shit
3	3	0	2	1	1	!!!!!!!!! RT @C_G_Anderson: @viva_based she look like a tranny
4	6	0	6	0	1	!!!!!!!!!!!!! RT @ShenikaRoberts: The shit you hear about me might be true or it might be faker than the bitch who told it to ya 
5	3	1	2	0	1	!!!!!!!!!!!!!!!!!!"@T_Madison_x: The shit just blows me..claim you so faithful and down for somebody but still fucking with hoes! 😂😂😂"
6	3	0	3	0	1	!!!!!!"@__BrighterDays: I can not just sit up and HATE on another bitch .. I got too much shit going on!"
7	3	0	3	0	1	!!!!“@selfiequeenbri: cause I'm tired of you big bitches coming for us skinny girls!!”
8	3	0	3	0	1	" & you might not get ya bitch back & thats that "
9	3	1	2	0	1	"
10	3	0	3	0	1	" Keeks is a bitch she curves everyone " lol I walked into a conversation like this. Smh
11	3	0	3	0	1	" Murda Gang bitch its Gang Land "
12	3	0	2	1	1	" So hoes that smoke are losers ? " yea ... go on IG
13	3	0	3	0	1	" bad bitches is the only thing that i like "
14	3	1	2	0	1	" bitch get up off me "
15	3	0	3	0	1	" bitch nigga miss me with it "
16	3	0	3	0	1	" bitch plz whatever "
17	3	1	2	0	1	" bitch who do you love "
18	3	0	3	0	1	" bitches get cut off everyday B "
19	3	0	3	0	1	" black bottle & a bad bitch "
20	3	0	3	0	1	" broke bitch cant tell me nothing "
21	3	0	3	0	1	" cancel that bitch like Nino "
22	3	0	3	0	1	" cant you see these hoes wont change "

➢Data cleaning

Data cleaning is one of the important parts of machine learning. It plays a significant part in building a model. It surely isn't the fanciest part of machine learning and at the same time, there aren't any hidden tricks or secrets to uncover. However, the success or failure of a project relies on proper data cleaning. Professional data scientists usually invest a very large portion of their time in this step because of the belief that "Better data beats fancier algorithms". If we have a well-cleaned dataset, there are chances that we can get achieve good results with simple algorithms also, which can prove very beneficial at times especially in terms of computation when the dataset size is large. Obviously, different types of data will require different types of cleaning. However, this systematic approach can always serve as a good starting point. The following are the most common steps involved in data cleaning: 1. Data inspection and exploration: This step involves understanding the data by inspecting its structure, identifying missing values, outliers, and inconsistencies. 2. Handling missing data: Missing data is a common issue in real-world datasets, and it can occur due to various reasons such as human errors, system failures, or data collection issues. Various techniques can be used to handle missing data, such as imputation, deletion, or substitution. 3. Handling outliers: Outliers are extreme values that deviate significantly from the majority of the data. They can negatively impact the analysis and model performance. Techniques such as clustering, interpolation, or transformation can be used to handle outliers. 4. Data transformation: Data transformation involves converting the data from one form to another to make it more suitable for analysis. Techniques such as normalization, scaling, or encoding can be used to transform the data. 5. Data integration: Data integration involves combining data from multiple sources into a single dataset to facilitate analysis. It involves handling inconsistencies, duplicates, and conflicts between the respective.

.

➢ ***code***

```
import pandas as pd
import numpy as np
import string
from sklearn.feature_extraction.text import CountVectorizer
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier
```

```
import nltk
import re
from nltk.corpus import stopwords
nltk.download('stopwords')
stopword=set(stopwords.words('english'))
stemmer = nltk.SnowballStemmer("english")
```

```
data = pd.read_csv("labeled_data.csv")
print(data.head())
```

```
data["labels"]=data["class"].map({0:"Hate speech",1: "offensive speech",
2:"No Hate and offensive speech"})
```

```
data=data[["tweet","labels"]]
```

```
data.head()
```

```
def clean(text):
    text = str(text).lower()
    text = re.sub('\[.*?\]', '', text)
    text = re.sub('https?://\S+|www\.\S+', '', text)
    text = re.sub('<.*?>+', '', text)
    text = re.sub('[%s]' % re.escape(string.punctuation), '', text)
    text = re.sub('\n', '', text)
    text = re.sub('\w*\d\w*', '', text)
    text = [word for word in text.split(' ') if word not in stopword]
    text=" ".join(text)
    text = [stemmer.stem(word) for word in text.split(' ')]
    text=" ".join(text)
    return text
```

```
data["tweet"] = data["tweet"].apply(clean)
```

```
x = np.array(data["tweet"])
y = np.array(data["labels"])
```

```
cv = CountVectorizer()
X = cv.fit_transform(x) # Fit the Data
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.33,
random_state=42)
```

```
model= DecisionTreeClassifier()
```

```
y_pred=model.predict(X_test)
```

```
from sklearn.metrics import accuracy_score
print(accuracy_score(y_test,y_pred))
```

```
i="you are too bad and i dont like your attitude"
i=cv.transform([i]).toarray()
print(model.predict((i)))
```

➢ conclusion

hate speech detection is a crucial task in the field of machine learning. The detection of hate speech aims to identify and classify offensive, discriminatory, or harmful language in various forms of text, such as social media posts, online comments, or public forums.

The goal is to develop models and algorithms that can automatically detect and flag such content, contributing to safer online environments and promoting inclusive and respectful communication. Advancements in machine learning techniques have facilitated the development of hate speech detection models. These models often leverage approaches such as text classification, sentiment analysis, and natural language processing techniques.

They are trained on labelled datasets that consist of examples of hate speech and non-hate speech instances, allowing them to learn patterns and characteristics of offensive language. However, hate speech detection remains a challenging task due to the complexity and evolving nature of language. Context, sarcasm, cultural references, and linguistic nuances can make it difficult to accurately classify certain instances of hate speech.

Additionally, biases and subjectivity in labelling data can impact the performance and fairness of the models.Our work has made several contributions to this problem. We introduced a method for automatically classifying hate speech.